Computational Snapshot Multispectral Cameras

- **全采样系统**
  - 滤光片光谱仪
    - rotating wheel
    - birefringent liquid crystal plates
    - 需要抗反射涂层以减少光通量耗散
  - 扫描式光谱仪
    - 推扫式(pushbroom)、掸扫式(whiskbroom)
    - 低时间分辨率，高空间分辨率
    - 具有一定机械复杂性和标定复杂度
  - 基于干涉的方法（傅里叶光谱仪）
    - 将图像通过干涉投影到传感器上，对应不同的颜色通道
    - 基于傅里叶变换
- **欠采样系统**
  - 种类
    - Prism-mask video imaging spectrometry(PMVIS) 棱镜掩膜视频光谱成像
      - 编码孔径为均匀分布
    - Single dispersive coded aperture snapshot imager(SD-CASSI) 单色散编码孔径快照式光谱成像
      - 应用压缩感知原理
      - 编码孔径为随机分布
    - Dual dispersive coded aperture snapshot imager(DD-CASSI) 双色散编码孔径快照式光谱成像
      - 采用两个色散编码孔径使最终数据立方体是unsheared
    - Spatial-spectral coded compressive spectral imager(SSCSI) 空间-光谱编码压缩光谱成像
      - 3-D-CASSI
      - 类似于DD-CASSI，但标定难度降低
    - 此后衍生出了 CCASSI(Colored~)
  - 压缩感知理论证明：感知矩阵 与 正交基矩阵 的相干性越低，所需采样数越少，采样效率越高

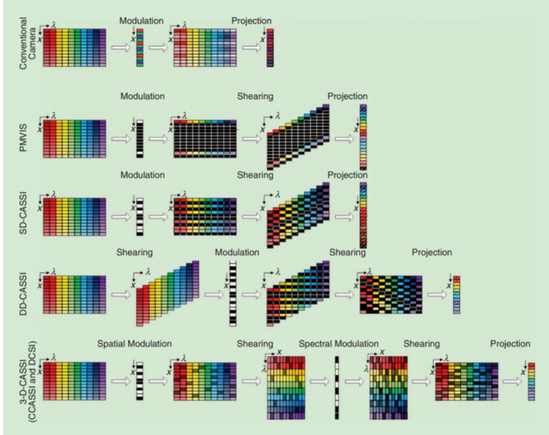

FIGURE 1. Sensing matrices of existing sampling schemes for multispectral acquisition. The spectral and spatial data matrix is high dimensional, and current camera sensors can capture only a low-dimensional projection of the spectral data. The projection process can be regarded as a sensing matrix for the high-dimensional spectral data cube, and state-of-the-art computational multispectral imaging methods can be summarized as different sensing matrices.

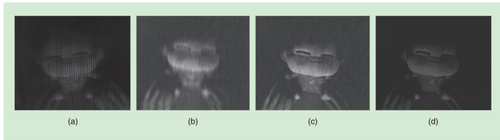

FIGURE 6. Measurement snapshots of the (a) PMVIS, (b) SD-CASSI, (c) DD-CASSI, and (d) 3-D-CASSI systems.

- **欠采样系统的评估比较**
  - 计算重建性能
    - 采用 PCA基 和 K-SVD基 作为稀疏基
    - 采用ADMM算法重建（PMVIS直接采用线性插值即可）

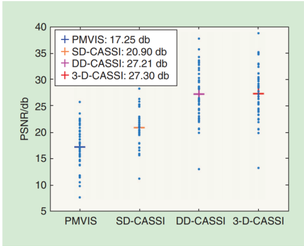

Table 1. The spectral sensing coherence information between the sensing matrices of different systems and the learned bases.

	SD-CASSI	DD-CASSI	3-D-CASSI	PMVIS
K-SVD	0.7920	0.7787	0.7737	0.8148
PCA	0.7048	0.6432	0.6663	0.7251

    - 结果显示 PMVIS效果最差，因为其极低的采样率，其他三种方法都可，DD-CASSI和3-D-CASSI最优
    - 随机的编码孔径并不是最优的
  - 光通量与系统复杂度

Table 2. Typical parameters for the four types of undersampling systems.

	SD-CASSI	DD-CASSI	3-D-CASSI	PMVIS
Light throughput	0.5	0.5	0.25	$\frac{1}{n}$
Number of optical elements	6	9	8	6

- **讨论&延申**
  - 视频速率多光谱相机 & 多次快照

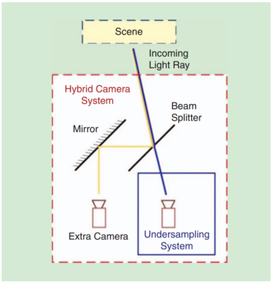

Table 3. PSNR comparison of three multisnapshot systems.

Systems	Hybrid PMVIS (+ Extra Camera) Simple Algorithm	Hybrid PMVIS (+ Extra Camera) Complex Algorithm	SD-CASSI (Measure Twice)	Hybrid CASSI (+ Extra Camera)
PSNR (DB)	25.86	33.19	28.04	32.10

  - 欠采样的其他高级理论
    - 低维线性子空间下的高斯混合模型(GMM)的应用
      - 协方差矩阵低秩